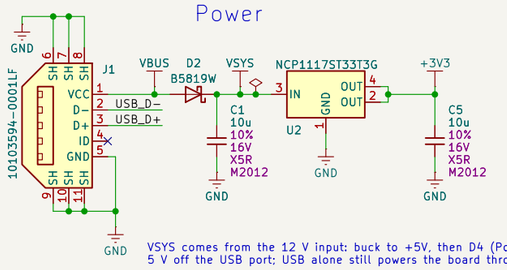
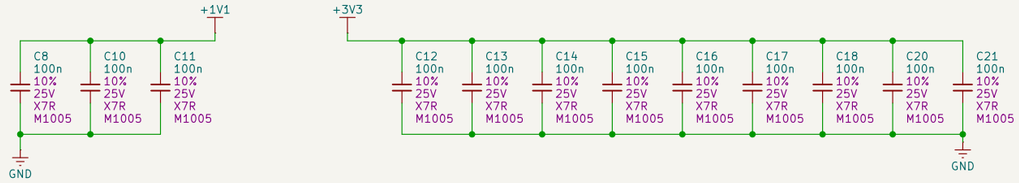
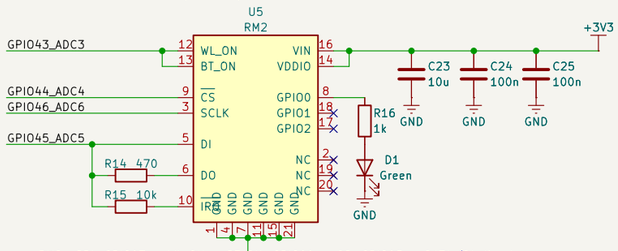
Power



VSYS comes from the 12 V input: buck to +5V, then D4 (Power sheet). D2 keeps that 5 V off the USB port; USB alone still powers the board through D2 (no +5V on the ribbons then).

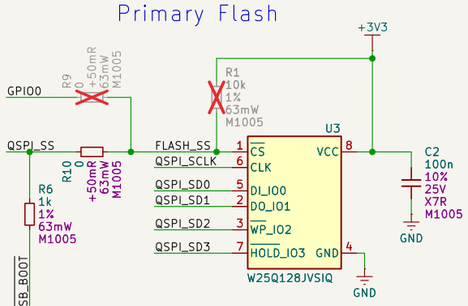


Radio (RM2)



RM2 gSPI: GPIO43 = WL_ON + BT_ON, GPIO44 = CS, GPIO45 = DATA (DI direct, DO through R14, IRQ through R15), GPIO46 = CLK. Same circuit as the Pico 2 W, moved off GPIO23/24/25/29; set CYW43_DEFAULT_PIN_* in the board header. Hand-solder U5 (not stocked at LCSC). Keep copper out from under the antenna end.

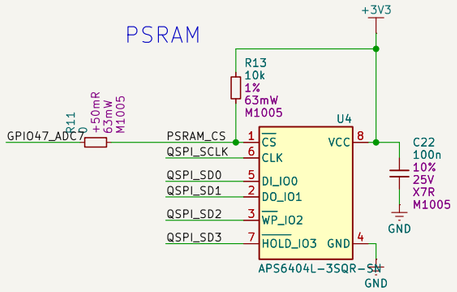
Primary Flash



If using RP2354 (internal flash), remove R10. R9 can be fit to use U3 as secondary memory.

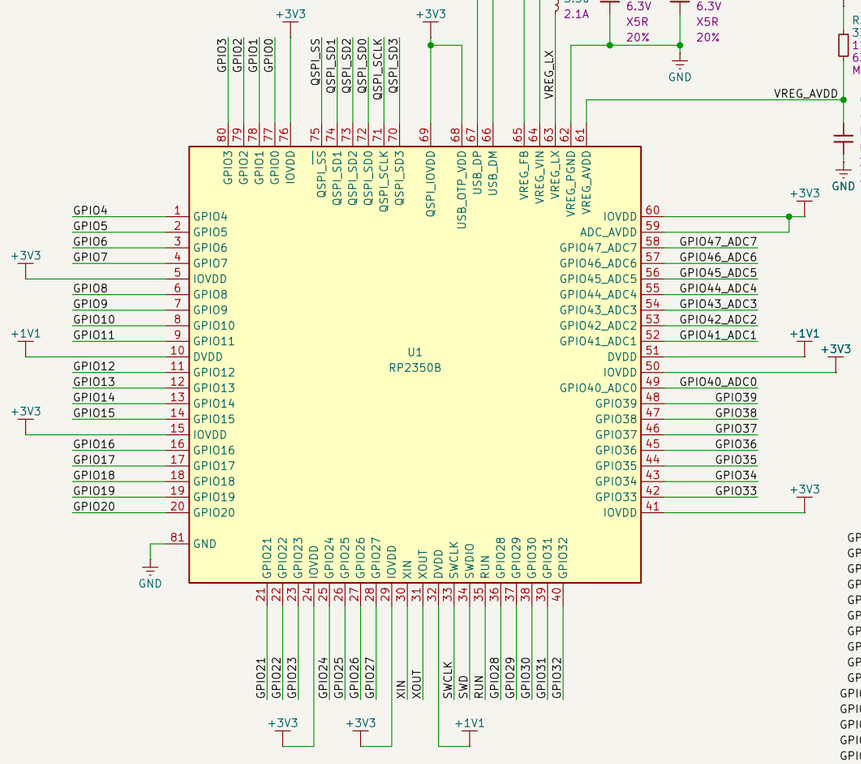
BOOTSEL Button

PSRAM



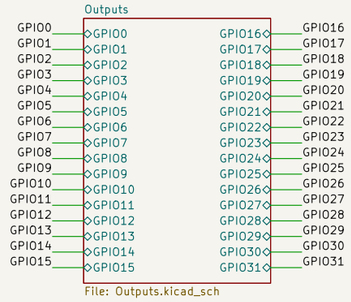
8 MB PSRAM (U4) on the second QSPI chip select (XIP_SS_N[1]), moved from GPIO0 to GPIO47. R13 pull-up keeps it deselected at power-up (GPIOs default to pull-downs). Fit R11 = 0R.

MH1 NPTH FD2.75mm SM3.15mm KO6mm
MH2 NPTH FD2.75mm SM3.15mm KO6mm
MH3 NPTH FD2.75mm SM3.15mm KO6mm
MH4 NPTH FD2.75mm SM3.15mm KO6mm



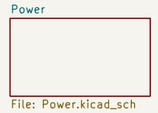
Make sure R7 and R8 are close to RP2350

GPIO0–31 to the four output shrouds



File: Outputs.kicad_sch

12 V in, 5 V buck



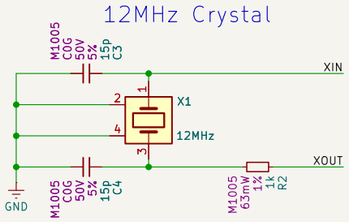
File: Power.kicad_sch

RTC and headers

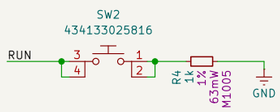
RTC: the DS3231 circuit from universal-strip-controller (U9), on I2C0: SDA = GPIO32, SCL = GPIO33, INT/SQW = GPIO34 (open drain, pulled up). 32KHZ and RST unused. CR1220 in a Keystone 3001 holder.

J2: spare GPIO35–42. UART1 TX/RX = GPIO36/37. GPIO40–42 = ADC0–2. J3: SWD in a Pico's debug pin order. GPIO0–31 go to the four output shrouds (Outputs sheet). GPIO43–47 are the RM2 and the PSRAM.

12MHz Crystal



RUN Button



Note: S1 schematic revision indicates that this is the 1st revision of the schematic/BOM associated with the R4 PCB. R4–S1 therefore uniquely identifies the fully built PCB Assembly (PCBA). This is a change from the previous revision numbering to align with the internal Raspberry Pi product revisioning scheme

VARIANT	VARIANT1
REF:	RPI-RP2350B-MINIMAL
DESCR:	RP2350B QFN-80 Minimal Design Example

SCHEMATIC/BOM REVISION: S1

BOM is autogenerated from schematic, therefore schematic rev. == BOM rev. Keep Titleblock rev. in sync with above. Reset to 1 when creating new R num.

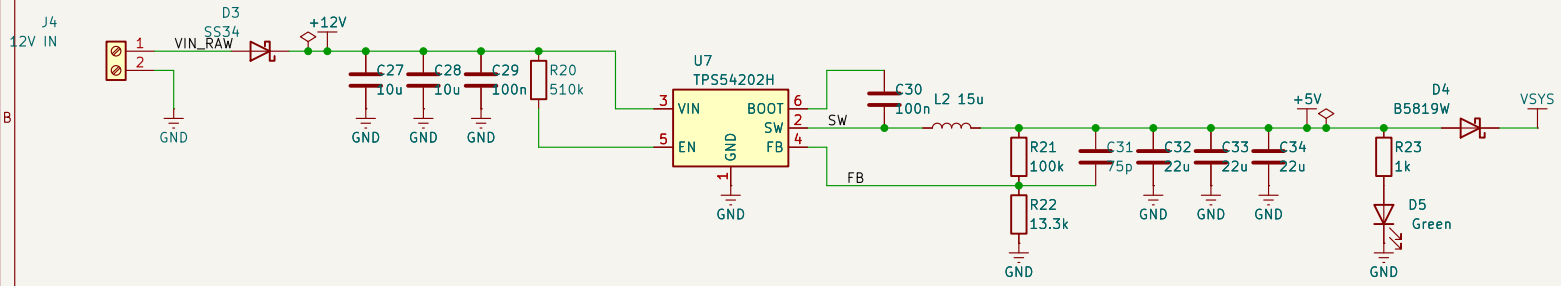
PCB/PWB REVISION: R4 <- PCB/PWB rev this schematic is for

MANUFACTURER: XXX
MANUF. PN: XXX
CIRCUITS/PANEL: XXX
DESCRIPTION: XXX

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TITLE	LED processor board (RP2350B + RM2)			
DATE	2026-10-14	REF	RPI-RP2350B-RM2 / RPI-RP2350B-MINIMAL	REV R4-S1
SIZE	A3	DRAWN	Original design: Mike Stimson, Raspberry Pi Ltd	SHEET 1/3

12 V in, 5 V buck



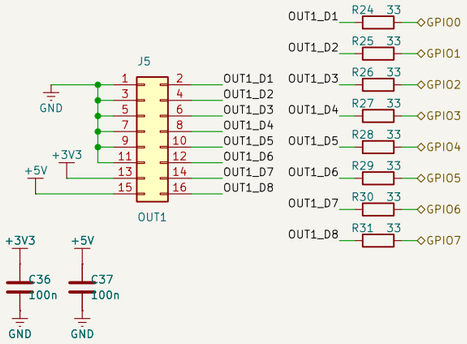
U7 values follow TI's TPS54202HEVM-716 5 V design (100k / 13.3k with 75 pF across the top resistor: 0.596 V x (1 + 100/13.3) = 5.08 V; 15 uH; 500 kHz). R20 enables U7 from the 12 V side. D3 blocks a reversed 12 V lead. The load is small: the 3.3 V regulator on VSYS (RP2350, RM2, flash, PSRAM) and the four output boards' shifters (VCCB on +5V, VCCA on +3V3), well under 0.5 A. +5V feeds the ribbons directly; VSYS gets it through D4 so USB can't back-feed the buck.

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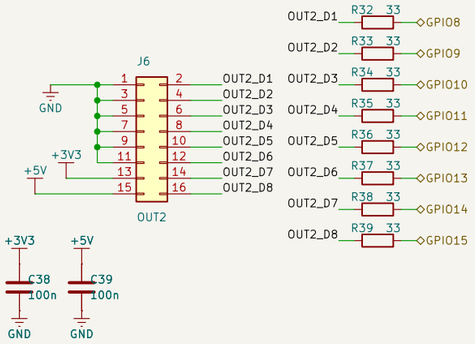
TITLE	LED processor board: Power				
DATE	2026-10-19	REF		REV	A
SIZE	A4	DRAWN		SHEET	2/3

Output shrouds: 16-pin IDC to the output boards. Pin 1 is the triangle on the shroud.
Odd pins 1-11 GND, 13 = +3V3 (shifter VCCA), 15 = +5V (shifter VCCB); data 1-8 on even pins 2-16
(same as led-output-board J2 and MegaTree docs/power.md). 33R series resistors at the RP2350 end.

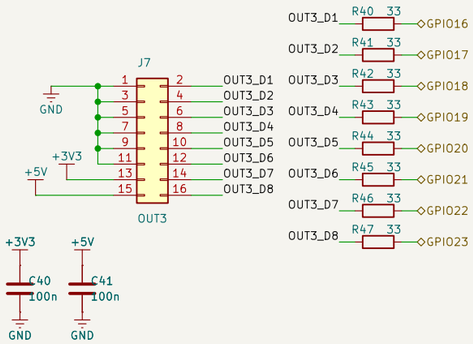
OUT1: GPIO0-7



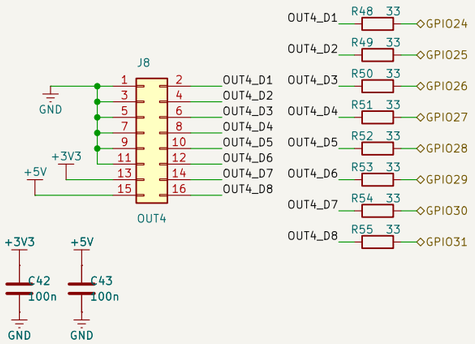
OUT2: GPIO8-15



OUT3: GPIO16-23



OUT4: GPIO24-31



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TITLE	LED processor board: Outputs				
DATE	2026-10-19	REF		REV	A
SIZE	A3	DRAWN		SHEET	3/3